

Needs and Learner Analysis

Enhancing Instructional Design Pedagogy for Online Technical and Professional Communication Education at the University of Arizona

This paper serves as a comprehensive needs and learner analysis for the instructional design project aimed at enhancing online Technical and Professional Communication (TPC) education within the University of Arizona's Writing Program. A needs analysis, as Stefaniak (2021) posits, "provides an opportunity for an instructional designer to develop instructional materials that can have a meaningful impact on their learning audience" (p. 54). Thus, this needs analysis discusses the setting for the analysis, the profile of the target learners, tools and approaches employed for the analysis, and the results that have implications for tailoring instructional design.

Setting for Implementation:

- The context for this analysis is the University of Arizona's Writing Program, which is undergoing substantial restructuring as it separates from the English department. The program's commitment to providing high-quality Professional and Technical Writing (PTW) courses remains unchanged, although it is adapting to new educational paradigms. This structural change emphasizes the urgency of aligning instructional design with evolving educational requirements. The Writing Program is a dynamic and critical environment for instructional design improvements, particularly as PTW undergoes a transformation from 300-level courses to a 100-level General Education (Gen Ed) format. The project's ultimate goal is to position the Writing Program as a leader in online PTW education and ensure its competitiveness within the changing educational landscape.

Identifying Target Learners:

- The primary target learners for this analysis are undergraduate students enrolled in online Technical and Professional Communication (TPC) courses at the University of Arizona. This diverse group varies in age, educational background, and technological proficiency. Their motivations, learning preferences, and technological literacy levels differ significantly, but they all share the common goal of gaining specialized knowledge and skills in professional and technical communication. This diverse range of learners is critical, particularly in the context of the transition of PTW to a certificate program in English and the potential shift to Gen Ed-level courses. The need to understand their information literacy needs, technological competencies, and learning preferences is essential for designing effective instructional approaches.

Identifying Tools and Approaches:

- A variety of tools and approaches will be employed to conduct the needs and learner analysis:
 1. **Surveys:** Prior technical writing Google Forms and Student Course Surveys were administered to learners to gather quantitative data about their backgrounds, prior experiences, technological competencies, and online learning preferences.
 2. **Interviews:** Qualitative insights will be collected through in-depth interviews with a select group of students. This approach allows for a more nuanced understanding of specific learner needs, challenges, and preferences.
 3. **Document Review:** Existing course materials, syllabi, and program documentation will be reviewed to evaluate the current state of instructional design, providing insights into strengths and weaknesses.
 4. **Learning Management System (LMS) Data:** Data from the University's LMS will be analyzed to gain insights into students' engagement with course materials, participation rates, and performance data. This data will be crucial for understanding how students interact with the course content and assessing the effectiveness of the current instructional design.

Results of Analysis:

- The analysis uncovers a range of insights into the characteristics, strengths, weaknesses, and potential areas of difficulty among the target learners. These findings are useful in informing recommendations for enhancing instructional design to align with their specific needs and preferences.

Target Learners:

- The target learners in TPC courses exhibit diverse backgrounds, technological proficiency, learning preferences, high motivation, and goal orientation. However, they differ significantly in terms of their prior experiences with technical and professional writing, information literacy, and online learning.

Strengths:

- While the learners in TPC courses exhibit diverse strengths, they all share high motivation and goal orientation. Moreover, many have technological proficiency.

Weaknesses:

- Some learners may lack experience in technical writing, requiring additional support in acquiring these skills. Additionally, challenges related to time management, self-regulated learning, and varying learning styles may pose obstacles.

Potential Areas of Difficulty:

- The transition to Gen Ed-level courses may be challenging for students accustomed to the previous 300-level PTW courses. The varying levels of information literacy among learners could affect their ability to locate, evaluate, and use information effectively. Furthermore, balancing coursework with personal and professional commitments is a common challenge.

Recommendations:

- Based on the analysis, several key recommendations are outlined to enhance instructional design:
 1. Develop modular course content to accommodate diverse learning styles.
 2. Provide additional support and resources for students with less experience in technical writing.
 3. Implement regular assessments and feedback mechanisms to address challenges in time management and self-regulated learning.
 4. Incorporate specific modules on information literacy to bridge gaps in this area.

These recommendations are instrumental in guiding the project to tailor instruction to the specific needs and characteristics of the target learners, ultimately enhancing the quality and effectiveness of PTW courses at the University of Arizona.

In conclusion, this needs and learner analysis provides a base for the subsequent phases of the proposed instructional design project. By understanding the characteristics, strengths, weaknesses, and potential areas of difficulty of the target learners, the project is able to develop instructional approaches that are responsive to their unique needs and preferences. Drawing from the principles outlined by Stratton (1976), this analysis ensures that the project aligns with well-established principles of effective communication and system design, ultimately enhancing the quality and effectiveness of PTW courses at the University of Arizona.

References

- Stratton, C. R. (1976). Needs Assessment for Communication System Design. *Journal of Technical and Professional Communication*, 6(2), 139-146.
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